

Analysis and Synthesis of Spatial Audio for VR Applications: Comparing SIRR and RSAO as Two Main Parametric Approaches

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ABSTRACT

In order to have a natural experience in a virtual reality environment, it is crucial to make the sound aligned with the surrounding room acoustics. The room acoustics are usually captured by room impulse responses (RIRs) which can be employed to regenerate the same audio perception. In this paper, we applied and compared two main parametric approaches, Spatial Impulse Response Rendering (SIRR) and Reverberant Spatial Audio Object (RSAO) to encode and render the RIRs. We showed that SIRR synthesizes the early reflections more precisely whereas RSAO is more accurate to render the late reverberation.

Index Terms: Virtual reality—Spatial Audio parametrization and rendering methods—Room impulse response

1 INTRODUCTION

In recent years, virtual reality (VR) has been employed for a wide range of applications from entertainment like video games and film industries, to healthcare, education and immersive communication. However, there has been considerably less studies on the audio perceptions compared to the vision aspect of the VR [1, 4, 5].

The main challenges for creating an immersive spatial audio experience are how to record the main features of the sound field easily, transmit them efficiently, edit them if required and reproduce the sound field with general rendering systems.

There are two major approaches that encode the audio scene with some attributes which can be tuned by the end users and rendered with arbitrary loudspeaker setups. The first one is Spatial Impulse Response Rendering (SIRR) [6, 7] that works in the time-frequency domain while the other method, Reverberant Spatial Audio Objects (RSAO) [2, 3], employs the time domain RIRs.

In this paper, we compare different properties of these approaches and discuss their strengths and weaknesses to apply for VR.

2 SIRR: ANALYSIS AND SYNTHESIS

In the Spatial Impulse Response Rendering (SIRR) [6, 7], the B-format signals are divided into windowed short time frames and each frame will be decomposed into frequency channels. Then the azimuth and elevation of the direct sound and early reflections along with the diffuseness which represents the reverberation, are calculated at each time-frequency (T-F) unit. Having the four channel B-Format signals $W(t)$, $X(t)$, $Y(t)$, and $Z(t)$, the vector $\mathbf{V}(t) = X(t)\mathbf{e}_x + Y(t)\mathbf{e}_y + Z(t)\mathbf{e}_z$, is defined where $\mathbf{e}_x$, $\mathbf{e}_y$, and $\mathbf{e}_z$ represent unit vectors in the direction of coordinate axes. By substituting the Fourier transforms of the B-format signals, the active intensity will be: $\mathbf{I}_a(\omega, m) = \frac{\sqrt{2}}{20} \text{Re}\{W^*(\omega, m)\mathbf{V}'(\omega, m)\}$, and the

diffuseness can be calculated as:

$$\psi(\omega, m) = 1 - \frac{\sqrt{2} \|\text{Re}\{W^*(\omega, m)\mathbf{V}'(\omega, m)\}\|}{|W(\omega, m)|^2 + |\mathbf{V}'(\omega, m)|^2/2}. \quad (1)$$

Having the intensity vector $\mathbf{I}_a$, the azimuth, $\theta(\omega, m)$ and the elevation $\phi(\omega, m)$ can be written as:

$$\theta(\omega, m) = \tan^{-1} \left[\frac{-\mathbf{I}_y(\omega, m)}{-\mathbf{I}_x(\omega, m)} \right], \quad (2)$$

$$\phi(\omega, m) = \tan^{-1} \left[\frac{-\mathbf{I}_z(\omega, m)}{\sqrt{\mathbf{I}_x^2(\omega, m) + \mathbf{I}_y^2(\omega, m)}} \right], \quad (3)$$

where $\mathbf{I}_x(\omega, m)$, $\mathbf{I}_y(\omega, m)$, and $\mathbf{I}_z(\omega, m)$ are the components of $\mathbf{I}_a$ in the directions of the corresponding Cartesian coordinate axes.

For the synthesis, the non-diffuse part is generated using vector based amplitude panning (VBAP) while the diffuse energy is distributed in a decorrelated form around the listener.

3 RSAO: ENCODING AND RENDERING

Reverberant Spatial Audio Object (RSAO) [2, 3] characterizes the RIRs with various parameters that can be sent to a renderer with an arbitrary loudspeaker setup. The main idea is to divide the RIR into three parts: direct sound, early reflections, and late reverberation.

For each reflection peak, four different parameters are calculated: time of arrival (ToA), direction of arrival (DoA), attenuation (level), and the spectrum modeled as 8 linear prediction coefficients (LPCs).

The parameters used to describe the late reverberation are the mixing time, T_m , the late peak and decay time constant. The mixing time was determined as the time when the echo density became greater than 1. Then an exponential curve was fitted to the RIR segment after the mixing time and its time constant was encoded for each octave sub-band as the late decay parameter. The VBAP method is also applied for the rendering.

4 SIRR AND RSAO COMPARISON

Early Reflections As mentioned in section 2, the DoAs of reflections at each time frame and frequency bin (azimuths and elevations) are calculated after extracting the non-diffuse energy of the signal using the SIRR approach. On the other hand, in section 3, ToA, DoA, attenuation (level) and spectrum of each early reflection as a peak in the time domain are estimated using the RSAO method.

When comparing these two approaches, the former one transforms the signal into time-frequency spectrum, whereas the latter one works with the signal in time domain. Therefore, as presented in [6], the early reflections are more distinguishable in the T-F domain as they are obviously frequency dependent. Hence, the DoAs are also different for each T-F unit, while in RSAO, DoAs are just related to the peaks in time.

The ToA and attenuation (level) of each reflection are somehow embedded in the SIRR algorithm as can be easily detected as onset times and amplitudes relative to the direct sound in the short time Fourier transform (STFT) spectrum.

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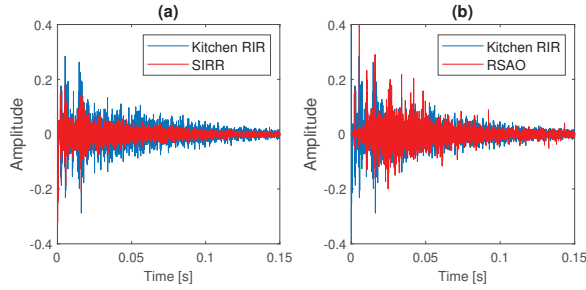


Figure 1: The recorded RIR of the Kitchen (blue) and the regenerated RIRs using SIRR (a) and RSAO (b).

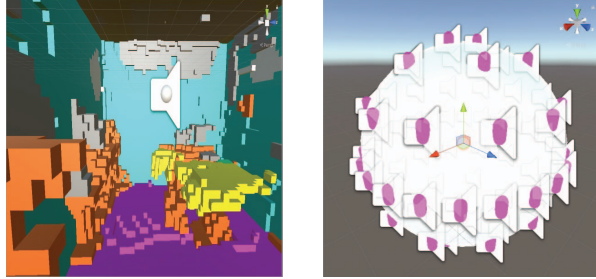


Figure 2: The Kitchen 3D model with one sound source (left) and the rendering 64 loudspeaker setup in Unity to regenerate the same acoustic properties of the room (right).

As stated earlier, the spectral properties of each reflection in RSAO is encoded using LPC coefficients, whereas SIRR works with Fourier transform as in STFT. The frequency components of the signal time frame in STFT is calculated with more coefficients, e.g. $N = 128$, compared to LPC, $N = 8$. Consequently, it can be concluded that SIRR conveys more precise information about the early reflections spectrum compared to RSAO.

Another advantages of using SIRR is that SIRR parameters can be extracted from any recorded signal whereas RSAO needs RIRs to estimate the parameters. However, RSAO is practically more efficient as it only sends the encoded parameters which reduces the size of data to be transferred.

Late Reverberation In the SIRR analysis a diffusion factor is calculated for each T-F unit which measures how scattered the sound energy is at that time and frequency in the recordings. Then for rendering implementation, this parameter is employed to extract the diffuse energy from the sound and then simulate the reverberation by diffusion convolution as a decorrelation technique. Even though this method aims at modelling the reverberation, the parameters such as mixing time, onset ramp length, late peak and exponential decay time constant which are estimated and employed by RSAO have not been adopted. For that reason, we can argue that RSAO models the late reverberation in a more accurate way compared to SIRR.

5 EXPERIMENT AND DISCUSSION

For this experiment, 3 different RIRs recorded by B-format microphones have been considered. The rooms are a Kitchen (KT), a Courtyard (CY), and a Meeting room (MR) [5]. Then SIRR and RSAO algorithms have been applied to estimate the RIR parameters for each room to be transmitted to a renderer.

For the rendering part, a virtual loudspeaker setup has been chosen with 64 loudspeakers positioned in elevations 0° , $\pm 28^\circ$, $\pm 56^\circ$, $\pm 80^\circ$. Then, given the transmitted RIR parameters, each method

Table 1: The metrics RT30 [s], C50 [dB], and EDT [s] for recorded ground truth RIR (GT) and rendered RIRs using SIRR and RSAO.

	Rooms	GT	SIRR	RSOA
RT60	KT	0.38	0.42	0.39
	CY	0.99	1.10	1.08
	MR	0.27	0.26	0.29
C50	KT	6.53	10.38	7.77
	CY	7.80	6.42	6.48
	MR	16.56	17.44	13.54
EDT	KT	0.42	0.25	0.39
	CY	0.93	1.20	0.29
	MR	0.17	0.20	0.27

has been employed to calculate the loudspeakers' feed. Having the loudspeaker feeds, a virtual environment is generated in the Unity platform including the Google Resonance Audio package with 64 loudspeaker around a sphere as shown in the right image in Fig. 2. The designated signals for each loudspeaker is then played simultaneously and the perceived regenerated RIR is recorded at the centre of the sphere in .ogg format with 4 channels. Fig. 1 compares the original recorded RIRs with the regenerated one in the Unity. As it can be seen SIRR has recovered the early reflections more precisely, while RSAO shows better performance for the late reverberation.

Different metrics have also been evaluated to compare the methods namely, the reverberation time (RT60), the early-to-late index (C50), and the early decay time (EDT). Table 1 shows the values for different rooms using different methods. Interestingly, each method gives better values for different metrics in different rooms. This is due to the fact that these metrics depend on both early reflections and late reverberation and their relative energy and not merely on each part of the RIR.

Since each method has some advantages and disadvantages, for the future work we propose a combined method which benefits from the positive aspects of each technique.

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